

25-11-05_update

Modify the wrapper function in ggclassify for blending color to accept any number of classes

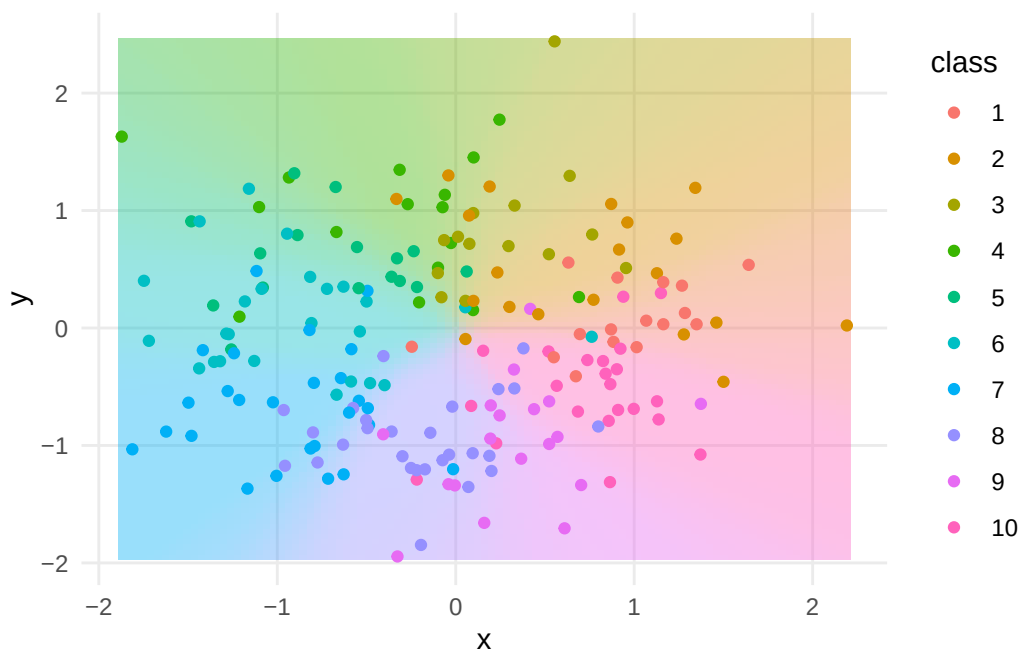
Blend style

returns the weighted average color of each point across all bootstraps.

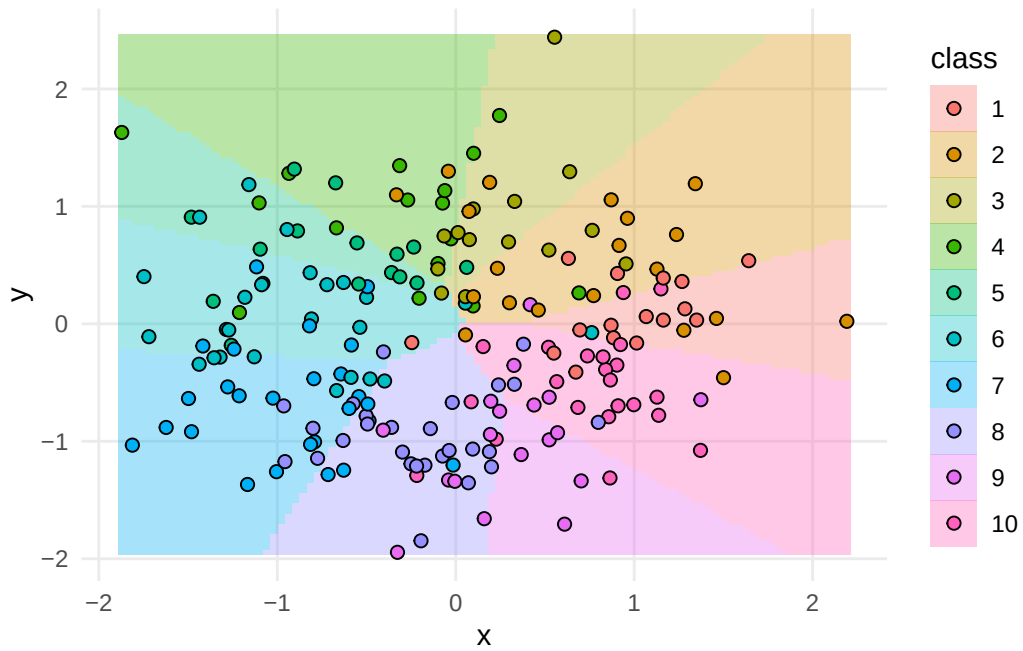
```
samp <- rnorm_mixture(200, mean = roots_of_unity(10), var = .2)
bidata <- tibble(x = samp[,1], y = samp[,2], class = attr(samp, "component"))

simpleplot4 <- ggplot(bidata, aes(x, y, fill = class))+
  geom_point_fill()

classify_boot_blend(bidata, simpleplot4, n = 100, res = 101, method = "lda")
```



```
ggplot(bidata, aes(x = x, y = y, fill = class))+
  geom_classify()+
  geom_point_fill()
```



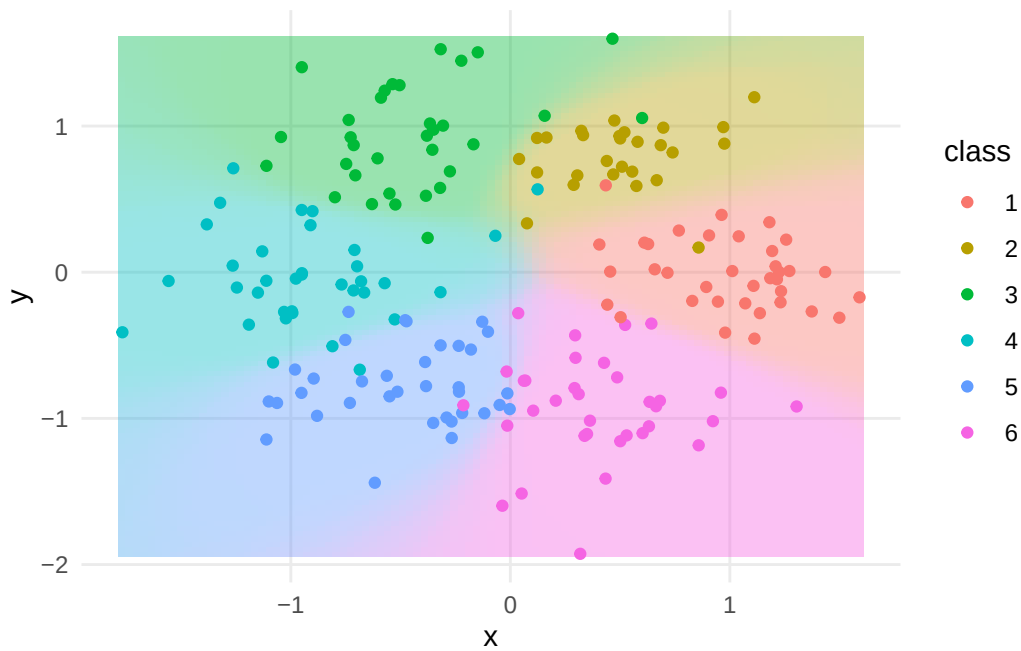
Default LDA of non-bootstrapped data for comparison

The same plots are repeated but with fewer classes, lower variability, and qda.

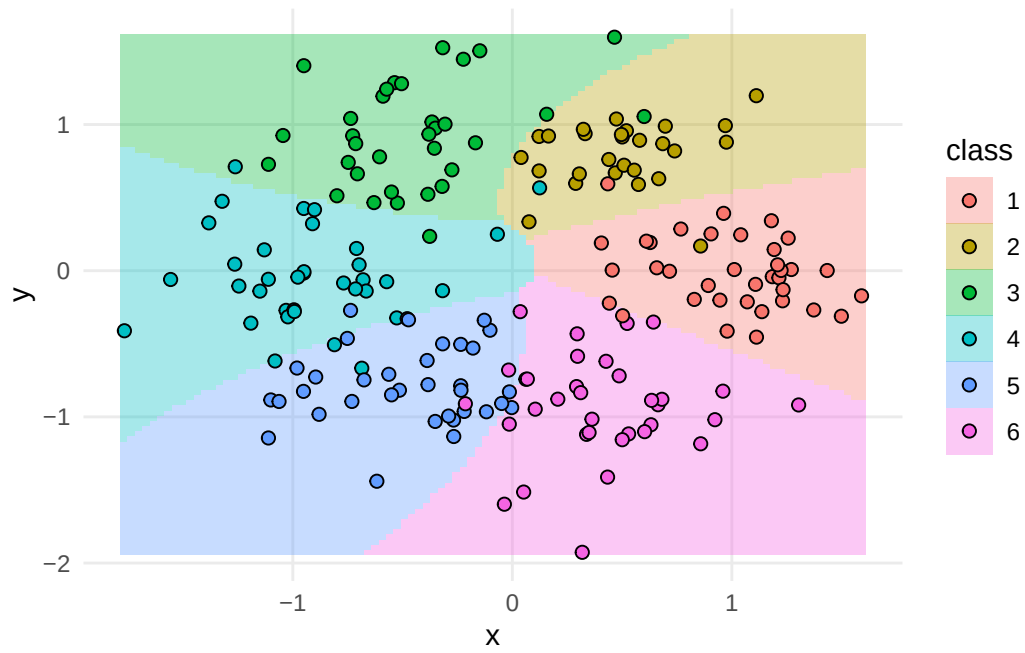
```
samp2 <- rnorm_mixture(200, mean = roots_of_unity(6), var = .1)
bidata2 <- tibble(x = samp2[,1], y = samp2[,2], class = attr(samp2, "component"))

simpleplot5 <- ggplot(bidata2, aes(x, y, fill = class))+
  geom_point_fill()

classify_boot_blend(bidata2, simpleplot5, n = 100, res = 101, method = "qda")
```



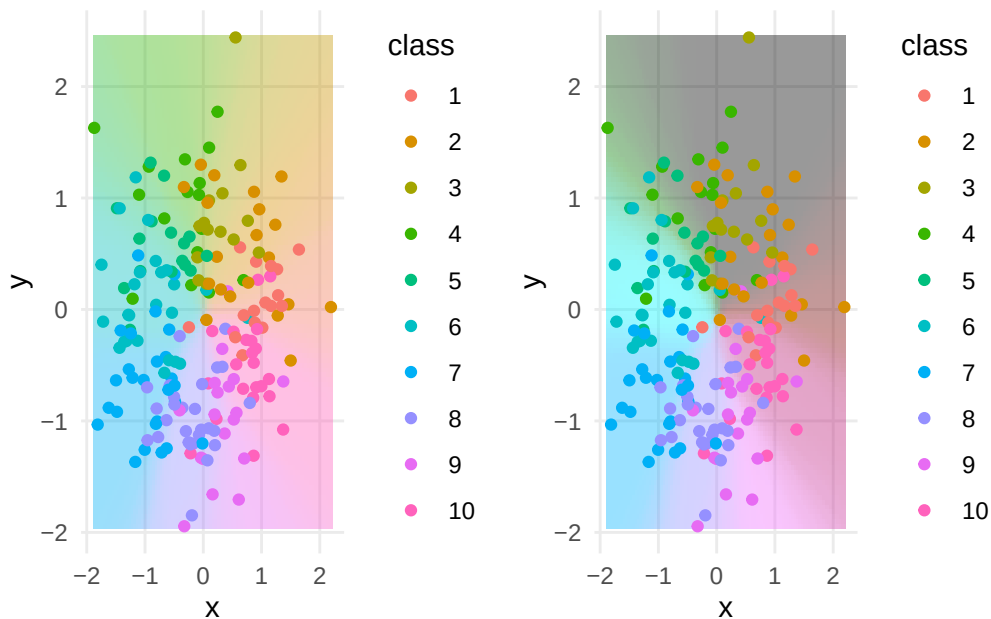
```
ggplot(bidata2, aes(x = x, y = y, fill = class))+
  geom_classify(method = "qda")+
  geom_point_fill()
```



Blend spectrum cmyk

As opposed to averaging the colors in the rgb space, i've also experimented with converting to cmyk, averaging there, and then converting back. This causes areas of uncertainty to trend towards black. I would say it was a failed idea, but I've shared it anyways.

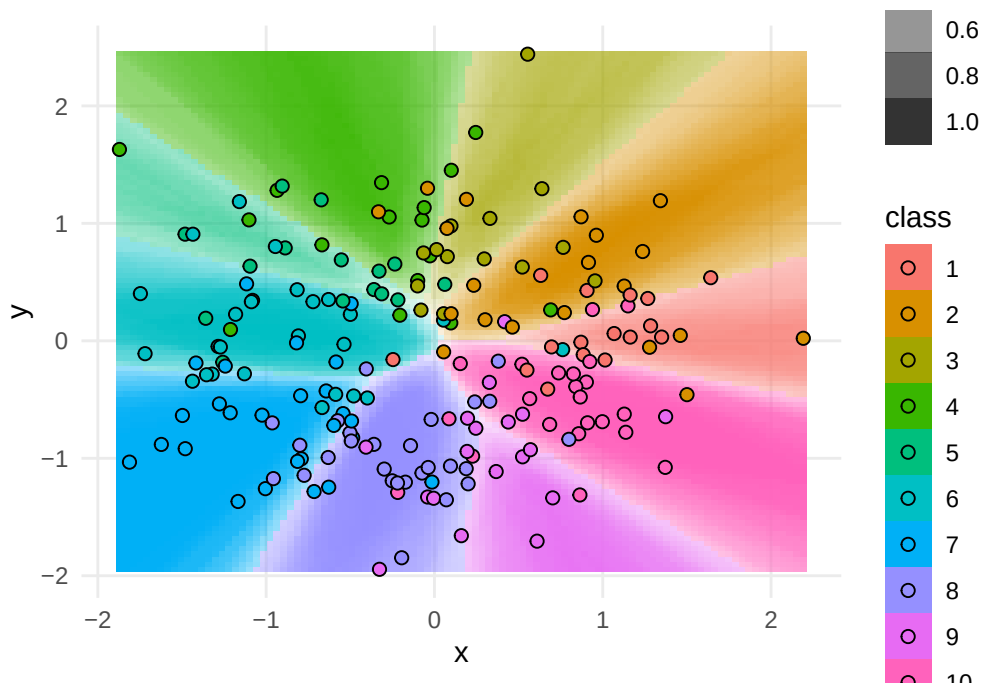
```
classify_boot_blend(bidata, simpleplot4, n = 100, res = 101)+  
classify_boot_blend(bidata, simpleplot4, n = 100, res = 101, spectrum = "cmyk")
```



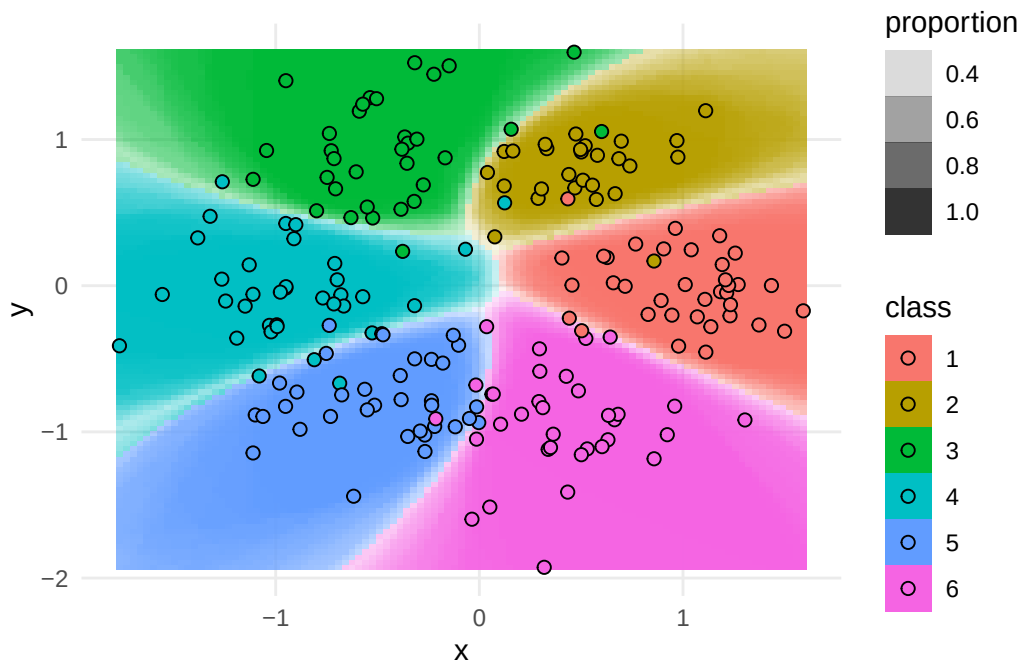
Fade style

As opposed to sharing the weighted average fill, you can instead present the fill of the majority class, and then alpha blend by what percent of the bootstraps have that majority. Name is currently pending at fade. Perhaps proportion would be better?

```
classify_boot_blend(bidata, simpleplot4, n = 100, res = 101, method = "lda", style = "fade")
```



```
classify_boot_blend(bidata2, simpleplot5, n = 100, res = 101, method = "qda", style = "fade")
```



Issue fixing

Issue 2

I've been working on resolving the issues for ggclassify on github. Issue 2 had already been started but i finished adding to documentation for vowel with a longer description of how they generated the data, and info on where it has been used.

Issue 4

For the issue of certain classifiers failing on non-factor input. Previously I had a CLI error that stopped the user and told them to supply a factor. I swapped it to automatically make the variable into a factor, but issue a CLI information warning that I have automatically converted their input.

Issue 5

I've been working on cleaning up the documentation to potentially merge class and class boundary. Ran into some issues along the way.